

Topological Group Actions on Fukaya Categories

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1 Goals

We would like to define topological group actions on categories, in the hope of working through the following example. Let X be a symplectic manifold with a hamiltonian G -action. Then there is a functor

$$\mathrm{Fuk}(T^*G) \rightarrow \mathrm{Fuk}(X \times \bar{X})$$

which gives rise to a topological group action of G on $\mathrm{Fuk}(X)$. We will begin by describing $\mathrm{Fuk}(X)$, in both the X compact and non-compact cases (since T^*G is manifestly non-compact). Then, we will define topological group actions and consider the case of hamiltonian G -spaces.

2 Fukaya Categories

2.1 Vanilla Fukaya Categories

We have already seen how quantum cohomology arises by considering physical A-models. We return to the case of a sigma model of maps $\Sigma \rightarrow X$ with X a compact symplectic manifold. We would now like to consider boundary conditions for these maps. I.e. we suppose Σ has boundary $\partial\Sigma \neq \emptyset$, and we constrain our path integral to those maps $\varphi : \Sigma \rightarrow X$ such that $\varphi(\partial\Sigma) = L \subset X$. We call such conditions **Dirichlet boundary conditions** or **D-branes** for short.

Remark 2.1. I want to get to the details of the topological action on $\mathrm{Fuk}(X)$, so I'm going to only give a very brief overview of the Fukaya category, with little motivation. We could have many lectures on Fukaya categories; in fact, we could spend a whole semester on this topic.

Physical considerations lead to the following facts about L . We call L an A-brane, i.e. a boundary condition for the A-model, if L is an equivalence class of lagrangians in X modulo hamiltonian isotopy equipped with a flat line bundle, have trivial Maslov class, and which satisfy an additional “brane condition” (physically, this is a “tadpole cancellation property”). The details of these conditions won't concern us; suffice it to say that L needs to be a lagrangian that has some extra structure.

$\mathrm{Fuk}(X)$ is supposed to be the assignment to the point in a 2-dimensional TFT called the A-model. The relation to Elliot's talk last week is that the assignment to the circle in this theory is exactly the quantum cohomology he introduced. From general TFT considerations explained by Chris in the first weeks, $Z(S^1) = \mathrm{HH}^*(\mathrm{Fuk}(X))$, where I've implicitly passed to the oriented theory. In particular, we should define the morphism spaces. We let $\mathrm{Hom}(L, L')$ be the **Floer complex** of the lagrangians L, L' . To compute it, we first deform L, L' via hamiltonian isotopy to a situation such that they only have transverse intersections. The complex will then be generated by the intersection points. Unfortunately, we can't end here, since the number of intersections is not invariant under hamiltonian isotopy. Thus, we need to implement corrections which account for such degeneracies. To do this, we implement **instanton corrections**. Equip X with

an almost-complex structure J , and we will count pseudo-holomorphic strips which fill disks in between intersections. These are maps $u : \mathbb{R} \times [0, 1] \rightarrow X$ such that $\bar{\partial}_J u = 0$, i.e.

$$\partial_s u + J(u) \partial_t u = 0,$$

with boundary conditions

$$\begin{aligned} u(s, 0) &\in L \text{ and } u(s, 1) \in L', \\ \lim_{s \rightarrow \infty} u(s, t) &= p, \quad \lim_{s \rightarrow -\infty} u(s, t) = q, \end{aligned}$$

along with a finite energy condition. In the case that our lagrangians did not intersect transversely, we add a deformation term:

$$\partial_s u + J(t, u) (\partial_t u - X_H(t, u)) = 0,$$

where X_H is the hamiltonian vector field generated by a hamiltonian $H \in C^\infty([0, 1] \times X)$. In this case, the Floer complex is then generated by flow lines $\gamma : [0, 1] \rightarrow X$ with $\dot{\gamma}(t) = X_H(t, \gamma(t))$ such that $\gamma(0) \in L$ and $\gamma(1) \in L'$. So generators are perturbed intersection points of L with L' , where the perturbation is given by time 1 flow generated by H . In either case, we then build a moduli space of such solutions $\mathcal{M}(p, q; [u], J)$, where $[u]$ is the Maslov index (fixed on the moduli space). We then define

$$\partial p = \sum_{q, [u] \mid \text{ind}[u]=1} |\mathcal{M}(p, q; [u], J)| T^{\omega([u])} q,$$

where $\omega([u]) = \int u^* \omega$. Here, we now divulge that the Floer complex should have been generated *over the Novikov ring* Λ , which is

$$\Lambda = \left\{ \sum_{i=0}^{\infty} a_i T^{\lambda_i} \mid a_i \in k, \lambda_i > 0, \lim_{i \rightarrow \infty} \lambda_i = \infty \right\}.$$

Remark 2.2. Note that because of the need to take hamiltonian deformations, there is a well-defined self-Floer homology $\text{Hom}(L, L) = \text{End}(L)$.

Remark 2.3. In the previous, we required X to be compact. There is additional care needed for the non-compact case.

2.2 Wrapped Fukaya Categories

Let X be a Liouville manifold, i.e. an exact symplectic manifold such that the Liouville vector field is complete and outward-pointing at infinity. We require that X contain a compact domain X^{in} with boundary a smooth hypersurface (which we denote by abuse of notation ∂X) on which $\alpha = \lambda|_{\partial X}$ is a contact form with Z positively transverse to ∂X and with no zeroes outside of X^{in} . We can thus use the flow of the Liouville vector field to write

$$X = X^{in} \cup_{\partial X} [1, \infty) \times \partial X.$$

If we denote by r the coordinate on $[1, \infty)$, the Liouville vector field is $Z = r \partial_r$ on $(1, \infty) \times \partial X$ and we have the exact symplectic form $\omega = d(r\alpha)$.

In this setting, we can now consider non-compact lagrangians which have some prescribed behavior at ∞ . We define the wrapped Fukaya category $\mathcal{W}(X)$ to have as objects exact lagrangians which are **conical** at infinity, i.e. invariant under the flow of Z outside of a compact subset and such that the exact 1-form $\lambda|_L$ vanishes outside of a compact set. I.e. if L is noncompact then at infinity it must be the cone $(1, \infty) \times \partial L$ over some legendrian $\partial L \subset \partial X$. Furthermore, the hamiltonian isotopies used to define Floer homology are also of a special type. Now, we only allow hamiltonians $H : X \rightarrow \mathbb{R}$ that are of the form $H = r^2$ outside of a compact subset of X , where $r \in (1, \infty)$ as before. Outside of a compact set, we thus have that $X_H = 2rR$, where R is the Reeb vector field of α . Indeed, recall that by definition, $R \in \ker(d\alpha)$ and $\alpha(R) = 1$. Since

$$\omega = d(r\alpha) = \alpha \wedge dr + r d\alpha$$

and

$$\begin{aligned} \iota_{X_H} \omega &= dH \\ &= 2r dr, \end{aligned}$$

we have

$$\iota_R \omega = dr,$$

so $X_H = 2rR$. Given two lagrangians, the **wrapped Floer complex** $\text{Hom}(L, L')$ is generated by time 1 trajectories of the flow of X_H which start on L and end on L' . In other words, in X^{in} this is perturbed intersections of L and L' , while outside of X^{in} this is given by Reeb chords of any length from ∂L to $\partial L'$, see Fig. 1 below. As before, the differential is given by counting pseudo-holomorphic strips which solve

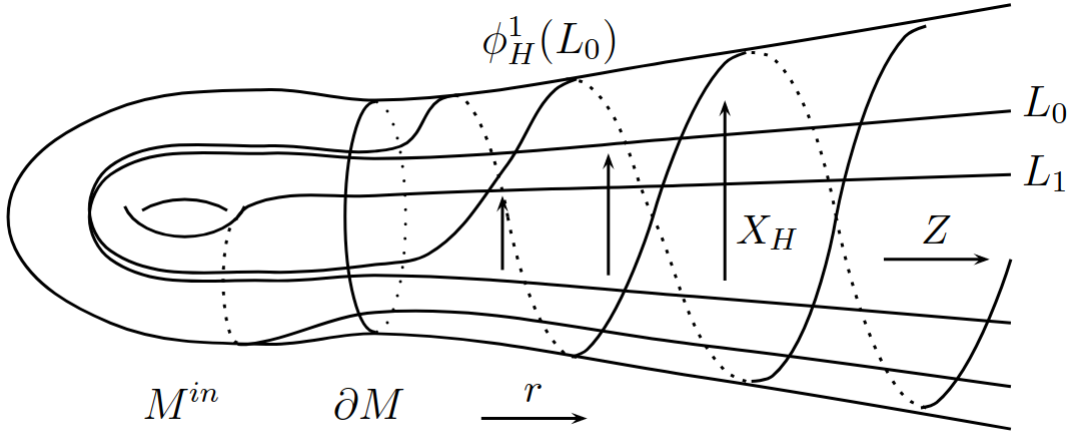


Figure 1: Here we see two lagrangians L_0 and L_1 in which we perturb L_0 by a time 1 hamiltonian isotopy $\phi_H^1(L_0)$.

Floer's equation with boundary on L and L' . Due to exactness of the lagrangians, we don't need to work over the Novikov ring Λ and can instead work over the ground field k .

3 Topological Group Actions

3.1 Some Homotopy Theory

Let G be a connected, compact Lie group. We will first investigate some homotopy properties of G . Note that since $G = B\Omega G$,

$$C^*(G) = C^*(B\Omega G).$$

Moreover, we have

Lemma 3.1.

$$C^*(G) = B(C_{-*}(\Omega G))^*,$$

where the RHS denotes the linear dual of the bar complex of cohomologically-graded chains on ΩG .

Proof. We simply write out the complex $C^*(B\Omega G)$ as

$$\lim \left(\cdots C^*(\Omega G) \otimes C^*(\Omega G) \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} C^*(\Omega G) \xleftarrow{\quad} C^*(\text{pt}) \right).$$

Similarly, the bar complex $B(C_{-*}(\Omega G))$ is resolution of $C_{-*}(\Omega G)$ by $C_{-*}(\Omega G) \otimes C_{-*}(\Omega G)$ -modules.

$$\cdots C_{-*}(\Omega G) \otimes C_{-*}(\Omega G) \otimes C_{-*}(\Omega G) \rightrightarrows C_{-*}(\Omega G) \otimes C_{-*}(\Omega G).$$

We can apply the functor $\text{Hom}_{C_{-*}(\Omega G) \otimes C_{-*}(\Omega G)}(-, k)$ to it, where we note that k is a $C_{-*}(\Omega G)$ -module because $C_{-*}(\Omega G)$ is a polynomial dga. This gives us the complex $C^*(B\Omega G)$. $\square$

Now, note that this bar resolution exactly computes $\text{Hom}_{C_{-*}(\Omega G)}(k, k) = C_{-*}(\Omega G)^!$, the Koszul dual of $C_{-*}(\Omega G)$. Thus, we find

Proposition 3.2. The Koszul dual of $C_{-*}(\Omega G)$ is $C^*(G)$.

Since $C^*(G)$ is an exterior algebra on finitely many generators by a theorem of Hopf, it follows that these algebras are Morita equivalent [BGS96]. We remark that $C_{-*}(\Omega G)$ is calculable in general.

Example 3.1. Let $G = S^1$. Then $\Omega G = K(\mathbb{Z}, 0) = \mathbb{Z}$, since $G = K(\mathbb{Z}, 1)$. Thus, $C_{-*}(\Omega G) = C_0\mathbb{Z} = k[\mathbb{Z}] = k[t, t^{-1}]$ is functions on $\mathbb{G}_m$.

Example 3.2. On the other hand, if G is simple, then $C^*(G)$ is known to be an exterior algebra on $\text{rk } G$ generators in odd degrees. In this case, $C_{-*}(\Omega G)$ will be a symmetric algebra on $\text{rk}(G)$ generators by Koszul duality.

General compact connected Lie groups are products of a simple group and a torus, so, in general,

$$C_{-*}(\Omega G) = k[u_1, \dots, u_k] \otimes k[t_1^\pm, \dots, t_n^\pm],$$

where $|t_i| = 0$ and the u_i have negative even degrees. If we want to calculate it, use Koszul duality to resolve k as a $C^*(G)$ -module via a Koszul resolution, use the fact that the Koszul resolution functor is monoidal, and compute.

3.2 Topological Group Actions

Recall that a TFT is a functor

$$Z : \text{Bord}_n \rightarrow \mathcal{C},$$

where $\mathcal{C}$ is an (∞, n) -category. In fact, Lurie's cobordism hypothesis is stated first for the **framed bordism category** $\text{Bord}_n^{\text{fr}}$. Since we are interested in 2d, given an m -manifold M of dimension ≤ 2 , a framing is by definition a trivialization

$$TM \oplus \underline{\mathbb{R}}^{2-m} \cong \underline{\mathbb{R}}^2,$$

where $\underline{\mathbb{R}}^k$ denotes the trivial rank k vector bundle. For example, in the case of a circle, this consists of a choice of a vector in the normal bundle and a choice of a tangent vector to S^1 . Now, given a topological space X and a rank n vector bundle V on X , we can consider manifolds with (X, V) -structure, i.e. manifolds which come equipped with a map $f : M \rightarrow X$ along with an isomorphism of vector bundles

$$TM \oplus \underline{\mathbb{R}}^{2-m} \cong f^*V.$$

Now, let $G \rightarrow O(n)$ be a group homomorphism, and consider the vector bundle $\mathbb{R}^n/G$ over BG determined by this homomorphism. We call $\text{Bord}_n^G = \text{Bord}_n^{(BG, \mathbb{R}^n/G)}$.

Example 3.3. There are equivalences

$$\text{Bord}_n^{\{1\}} \cong \text{Bord}_n^{\text{fr}} \quad \text{Bord}_n^{SO(n)} \cong \text{Bord}_n^{\text{or}} \quad \text{Bord}_n^{O(n)} \cong \text{Bord}_n.$$

The cobordism hypothesis for bordisms with G -structure roughly states that such theories are equivalent to functors such that $Z(\text{pt}^+)$ is a fixed point for the G -action in the target 2-category. To ask that the group action is **topological** is by definition to require a map from $C_{-*}(\Omega G)$ to the Hochschild cohomology of $\mathcal{C}$.

Remark 3.1. Both the Hochschild cohomology and $C_{-*}(\Omega G)$ are E_2 -algebras, and we require that the map in fact be an E_2 -algebra homomorphism.

Given a category $\mathcal{C}$, its Hochschild cohomology $\text{HH}^*(\mathcal{C})$ is defined as

$$\text{HH}^*(\mathcal{C}) = \text{End}_{\text{Fun}(\mathcal{C}, \mathcal{C})}(\text{id}_{\mathcal{C}}).$$

We remark that the zeroth Hochschild cohomology acts on the category. Indeed, given $\varphi \in \text{HH}^*(\mathcal{C})$ and $c \in \mathcal{C}$, since φ is a natural transformation of $\text{id}_{\mathcal{C}}$, it induces a map $\varphi_c : c \rightarrow c$. This is also the assignment (after twisting once by the Serre functor) to the circle in a 2d TFT with value on a point $\mathcal{C}$.

Remark 3.2. This is the derived enhancement of the statement that the true endomorphisms (rather than the derived endomorphism) act on the category $\mathcal{C}$.

4 Topological Group Actions from Hamiltonian G -spaces

Let X be a hamiltonian G -space, not necessarily compact. We ask that X be a smooth symplectic manifold. We will construct a domain wall

$$\text{Fuk}(T^*G) \rightarrow \text{Fuk}(\bar{X} \times X)$$

which will give rise to the desired hamiltonian G -action. Note first that by a theorem of Abouzaid, $\text{Fuk}(T^*G) \cong C_{-*}(\Omega G)\text{-mod}$, where $\text{Fuk}(M)$ will always denote the wrapped Fukaya category if M is not compact.

Recall that a **lagrangian correspondence** is a lagrangian $C \subset \bar{Y} \times Z$, which we view as a map in the symplectic category from $Y \rightarrow Z$. Here, $\bar{Y}$ denotes Y with the symplectic form $-\omega_Y$. The issue with the symplectic category is that in general it is not possible to compose lagrangian correspondences. Given two correspondences $C_1 \subset \bar{Y} \times Z_1$ and $C_2 \subset \bar{Z}_1 \times Z_2$, we would like to define their composition as $C_1 \times_{Z_1} C_2$, but this fails to be lagrangian unless we place some assumptions on C_1, C_2 .

Define a **generalized lagrangian correspondence** from Y to Z as a sequence $\underline{L} = \{L_{i,i+1} \subset \bar{A}_i \times A_{i+1}\}_{i=0}^{k-1}$, where $A_0 = Y$ and $A_k = Z$. Note that given generalized lagrangian correspondences $\underline{L}_{12}$ from Z_1 to Z_2 and $\underline{L}_{23}$ from Z_2 to Z_3 , we can define their composition as $(\underline{L}_{12}, \underline{L}_{23})$. This gets around the problem of them not having a *geometric* composition.

We call a symplectic manifold Z **monotone** if $c_1(Z) = \tau[\omega]$, where ω is the symplectic form on Z . Similarly, we call a lagrangian $L \subset Z$ **monotone** if there is a $\tau \geq 0$ such that for every smooth map $u : (D, \partial D) \rightarrow (Z, L)$, we have

$$\int_D u^* \omega = \tau \mu(u),$$

where μ is the Maslov index of μ . Since exact lagrangians have $\tau = 0$. Now, fix a Liouville manifold W with $2c_1(W) = 0$, a simply-connected, compact, monotone symplectic manifold Z with monotonicity constant τ , and a compact, τ -monotone lagrangian brane $C \subset \bar{W} \times Z$. We define a category $\text{Fuk}_C(Z)$ to have as objects either lagrangian branes $M \subset Z$ or generalized lagrangian correspondences (L, C) , where now $L \subset W$ is an exact, Maslov zero lagrangian brane in $\text{Fuk}(W)$. There is also an additional “admissibility” condition on the objects, which we omit. There are four types of morphism spaces, given by the four different kinds of pairings of objects. This will not be important for us, but they are

$$\begin{aligned} \text{Hom}(M_1, M_2) &= \text{Hom}_{\text{Fuk}(Z)}(M_1, M_2) \\ \text{Hom}(M, (L, C)) &= \text{Hom}_{\text{Fuk}(\bar{W} \times Z)}(L \times M, C) \\ \text{Hom}((L, C), M) &= \text{Hom}_{\text{Fuk}(\bar{Z} \times W)}(M \times L, \bar{C}) \\ \text{Hom}((L_1, C), (L_2, C)) &= \text{Hom}_{\text{Fuk}(W \times \bar{Z} \times Z \times \bar{W})}(L_1 \times \Delta_Z \times L_2, C \times \bar{C}). \end{aligned}$$

We then have

Theorem 4.1 ([EL18]). There is a functor $\Phi(C) : \mathcal{B} \rightarrow \text{Fuk}_C(Z)$ from the subcategory $\mathcal{B} \subset \text{Fuk}(Z)$ consisting of objects L such that (L, C) is an object of $\text{Fuk}_C(Z)$, and it acts by $\Phi(C)(L) = (L, C)$. Moreover, if (L, C) is geometrically composable, then $(L, C) \cong L \circ C$.

Let (X, ω) be a simply-connected monotone symplectic manifold with a hamiltonian G -action and moment map $X \rightarrow \mathfrak{g}^*$. We define the **moment correspondence** as the lagrangian submanifold

$$C = \{(g, \mu(gx), x, gx \in \overline{T^*G} \times \bar{X} \times X\}.$$

It is a fact that this correspondence is monotone, and hence by the theorem it defines a functor

$$\Phi(C) : \mathcal{B} \rightarrow \text{Fuk}_C(\bar{X} \times X)$$

from the subcategory of $\text{Fuk}(T^*G)$ consisting of exact lagrangian branes L such that (L, C) is a brane in the target. We then have

Proposition 4.2. The generalized lagrangian correspondence (T_1^*G, C) is composable and its geometric composition is the lagrangian $\Delta \subset \bar{X} \times X$.

Proof. The transverse intersection $(T_1^*G \times \bar{X} \times X) \cap C = \{1, \mu(x), x, x\}$, and the projection to $\bar{X} \times X$ is exactly the diagonal. $\square$

We now use

Theorem 4.3 ([Abo11]). The Fukaya category $\text{Fuk}(T^*G)$ is generated by a cotangent fiber. In particular, $\text{Fuk}(T^*G) = \langle T_1^*G \rangle$.

Thus, by restricting to the subcategory of $\mathcal{B}$ generated by T_1^*G , we can assume that it lands in $\text{Fuk}(\bar{X} \times X)$. By passing to derived categories, we thus have a functor

$$\text{Fuk}(T^*G) \rightarrow \text{Fuk}(\bar{X} \times X).$$

Furthermore, we have

Theorem 4.4 ([Abo10]). The category

$$\text{Fuk}(T^*G) = C_{-*}(\Omega G)\text{-mod}.$$

Thus, its Hochschild homology is exactly $\text{HH}_*(C_{-*}(\Omega G))$. Since this is a polynomial algebra on negative even and degree 0 generators, its Hochschild homology is this same algebra tensored with the algebra on the same generators x_i placed in degree $|x_i| - 1$. On the other hand, we have

Corollary 4.5. The Hochschild cohomology of $\text{Fuk}(X)$ is

$$\text{HH}^*(\text{Fuk}(X)) = \text{End}_{\text{Fun}(\text{Fuk}(X), \text{Fuk}(X))}(\text{id}) = \text{End}_{\text{Fuk}(\bar{X} \times X)}(\Delta).$$

Proof. The first equality is by definition. The second follows by the association to Δ of a functor $\Phi(\Delta) : \text{Fuk}(X) \rightarrow \text{Fuk}(X)$. By definition, this sends $L \mapsto (L, \Delta) = L \times_X \Delta = L$. Hence, $\Phi(\Delta) = \text{id}_{\text{Fuk}(X)}$. $\square$

Remark 4.1. The assignment $C \mapsto \Phi(C)$ defines an equivalence

$$\text{Fun}(\text{Fuk}(X), \text{Fuk}(X)) \cong \text{Fuk}(\bar{X} \times X).$$

I think this means that the Hochschild homology of $\text{Fuk}(\bar{X} \times X)$ is $\text{HH}^*(\text{Fuk}(X))^{\otimes 2}$, and that the even part of the algebra gets sent to exactly one copy of this, but I'm not sure how to formulate this correctly.

Example 4.1 (T^*S^1 and Mirror Symmetry [Aur13]). Consider the wrapped Fukaya category $\text{Fuk}(T^*G)$ for $G = S^1$. By Abouzaid's theorem, this is the same as $C_{-*}(\Omega S^1)\text{-mod} = k[t, t^{-1}]\text{-mod}$. Model T^*G as $\mathbb{R} \times S^1$ equipped with the standard Liouville form $rd\theta$ and the wrapping hamiltonian $H = r^2$. Consider the exact lagrangian submanifold $L = \mathbb{R} \times \text{pt}$. For the time-1 hamiltonian flow $\phi_H^1(L)$, we can label the intersection points of $\phi_H^1(L)$ with L by integers in increasing order along the real axis. The Floer differential vanishes identically since there are no rigid pseudo-holomorphic strips with boundary on L and $\phi_H^1(L)$. It's not hard to show that the product structure on the Floer complex is $x_i x_j = x_{i+j}$. Hence, the Floer complex is quasi-isomorphic to $k[t, t^{-1}]$, which explicitly shows Abouzaid's theorem. On the other hand, the mirror to $\mathbb{C}^* = T^*S^1$ is itself, and coherent sheaves on $\mathbb{C}^*$ is exactly $k[t, t^{-1}]\text{-modules}$.

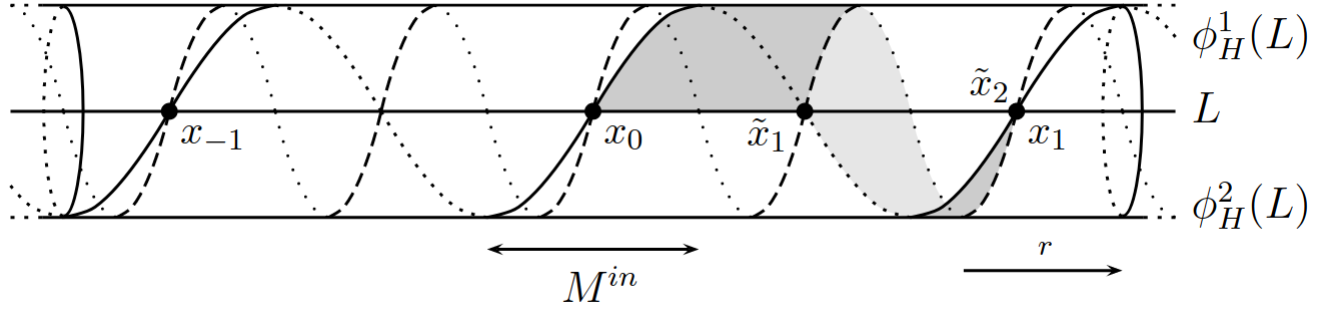


Figure 2: We view x_0 as the intersection point of L with a small pushoff of itself (say at $0 \in \mathbb{R}$). The other points are Reeb chords from $\partial L = \text{pt} \sqcup \text{pt}$ in the contact manifold $\partial M = S^1 \sqcup S^1$.

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